

Power BI Dashboard Analysis

1. How does customer distribution vary across different countries or cities?

The customer base is distributed across multiple countries and cities, with the highest concentration of customers located in the United States, Germany, and Austria. Major metropolitan cities contribute a larger share of customers compared to smaller cities. This distribution indicates that business operations are strongly concentrated in a few key geographic markets.

Visualization Used: Bar Chart / Map

2. What is the trend in customer orders over time?

Customer orders show a generally increasing trend over time with periodic fluctuations. Order activity increases during certain months, indicating seasonal demand patterns. The upward trend suggests growing customer engagement and business expansion.

Visualization Used: Line Chart

3. What is the distribution of customers by Contact Title or Region?

Customers are distributed across various contact titles such as Owner, Sales Representative, Marketing Manager, and Purchasing Manager. The majority of customers belong to managerial and purchasing roles. Regional analysis shows a concentration of customers in North America and Europe.

Visualization Used: Pie Chart / Stacked Bar Chart

4. How does order volume change over time?

Order volume demonstrates consistent growth throughout the analyzed period. Certain months experience higher order counts than others, suggesting seasonal variations and changing purchasing behavior. Peak periods contribute significantly to annual sales performance.

Visualization Used: Line Chart

5. What is the distribution of order values?

Most orders fall within the low-to-medium value range, while a smaller number of orders generate exceptionally high revenues. This distribution is positively skewed, indicating that a limited number of high-value orders contribute significantly to total sales.

Visualization Used: Histogram

6. What is the average order shipping duration?

The average shipping duration remains relatively stable across orders. Most orders are shipped within a few days of placement, indicating efficient logistics operations. A small number of orders experienced longer shipping durations due to operational or regional factors.

Visualization Used: Bar Chart / Box Plot

7. What is the count of employees by job title or region?

Sales Representatives constitute the largest employee group within the organization. Other employee categories include Vice President, Sales Manager, and other managerial roles. Employee distribution is concentrated within a limited number of operational regions.

Visualization Used: Tree Map / Stacked Bar Chart

8. What is the distribution of employee tenure?

Employee tenure varies significantly across the workforce. Several employees have long service durations, indicating organizational stability, while newer employees represent recent workforce expansion. The distribution suggests a balanced mix of experienced and newer staff members.

Visualization Used: Histogram

9. What is the reporting structure among employees?

The organizational hierarchy follows a structured reporting model where employees report to supervisors and managers. Senior management positions oversee operational and sales teams, ensuring effective coordination and decision-making throughout the organization.

Visualization Used: Hierarchical Tree / Org Chart

10. Which products have the highest sales volume?

A small number of products account for a significant proportion of total sales volume. These products consistently outperform others due to strong customer demand and frequent ordering patterns. They represent key revenue drivers for the business.

Visualization Used: Bar Chart / Tree Map

11. How does the sales volume vary across different product categories?

Sales volume differs substantially among product categories. Beverages generate the highest sales volume, followed by Dairy Products and Confections. Some specialized categories contribute lower sales volumes but may maintain higher profit margins.

Visualization Used: Tree Map

12. Can we visualize the pricing distribution of products?

Product pricing varies across categories and suppliers. Most products are concentrated within moderate price ranges, while a few premium products command significantly higher prices. This variation highlights the presence of both mass-market and premium offerings.

Visualization Used: Histogram / Box Plot

13. How many products are supplied by each supplier?

Supplier contribution varies considerably. Some suppliers provide a large portfolio of products across multiple categories, while others specialize in a limited range of products. Supplier diversity helps ensure product availability and supply chain stability.

Visualization Used: Bar Chart

14. How does product pricing vary across different suppliers?

Product prices differ among suppliers due to sourcing costs, product quality, and regional market conditions. Certain suppliers consistently provide premium-priced products, while others focus on cost-effective offerings. These differences directly influence profitability and purchasing decisions.

Visualization Used: Box Plot / Stacked Column Chart

15. What is the geographical distribution of suppliers?

Suppliers are geographically distributed across North America, Europe, and other international regions. European suppliers represent a significant share of the supplier network, reflecting the company's global sourcing strategy. Geographic diversity reduces dependency on a single market and strengthens supply chain resilience.

Visualization Used: Map Chart / Bubble Map